

Saurabh Kumar

saurabhbr2.0@gmail.com · +91-8252697270 · linkedin.com/in/saurabh-bharti-br09 · github.com/Srv108

PROFESSIONAL SUMMARY

Backend / Full-Stack Engineer experienced in building **scalable REST APIs**, real-time web applications, and **AI-powered RAG pipelines** using **Node.js, Next.js, ASP.NET Core, and TypeScript**. Proficient in **LangChain, MongoDB, PostgreSQL, JWT/OAuth 2.0, WebSockets, CI/CD**, and **Vector Search** — with a track record of shipping production-ready, performance-optimized software.

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, C#, C++, Java, Python

Frontend: React.js, Next.js 16, Tailwind CSS, HTML5, CSS3, Responsive Web Design

Backend: Node.js, Express.js, Fastify, ASP.NET Core Web API, REST API, WebSockets, Microservices, Entity Framework Core

AI / ML: LangChain.js, LangGraph, RAG (Retrieval-Augmented Generation), Vector Embeddings, OpenAI API, Anthropic Claude API, Generative AI

Databases: MongoDB, MongoDB Atlas Vector Search, PostgreSQL, MySQL, ORM, LINQ, Query Optimization, Indexing

Tools & DevOps: Git, GitHub, Docker, AWS (EC2, S3), Cloudinary, GitHub Actions, CI/CD, Turborepo, Postman, Swagger, Render

Concepts: JWT Authentication, OAuth 2.0, System Design, MVC, Repository Pattern, Factory Pattern, Dependency Injection, OOP, DSA, Agile/Scrum

PROFESSIONAL EXPERIENCE

.NET Developer — Programmer Analyst | Cognizant Technology Solutions Feb 2026 – Present

- Built a full-stack **Learning Management & Student Progress Tracking System** using **ASP.NET Core MVC** with role-based modules for Students, Teachers, and Admins, supporting enrollment, assignments, and analytics workflows.
- Designed and exposed **10+ RESTful APIs** covering enrollment, assignments, exams, attendance, and grade analytics using **MVC (Controller–Service–Model)** architecture for clean separation of concerns.
- Managed entire database lifecycle with **Entity Framework Core** (Code-First + Migrations) — schema design, seeding, and **zero-downtime versioned migrations** on MySQL and PostgreSQL.
- Optimized data access with **LINQ** queries, reducing boilerplate by **~40%** and improving code readability across all service layers.

Backend Developer Intern | Dehix (Freelancing Marketplace Startup) Aug 2025

- Built and shipped **scalable backend Fastify APIs** for a live freelancing marketplace supporting project posting, bidding, and profile management.
- Designed flexible **MongoDB schemas** handling dynamic project data (budgets, skill tiers, experience levels) and implemented validation & business-logic middleware.
- Iterated on API contracts with the frontend team, improving response shapes and reducing average round-trip latency by **~30%**.

PROJECTS

BrainHistory — AI-Powered Personal Knowledge Library 2025

Next.js 16, React 19, TypeScript, MongoDB Atlas Vector Search, LangChain.js, LangGraph, OpenAI, Anthropic Claude, NextAuth v5, Cloudinary, Tailwind CSS, REST API, Render

- Architected a **production-grade full-stack AI knowledge platform** enabling users to save, semantically index, and conversationally query 15+ content types — YouTube, Spotify, PDFs, GitHub, Reddit, Twitter/X, audio, and rich-text notes — via URL ingestion or file upload.

- Engineered an end-to-end **RAG pipeline** — content parsed via Cheerio, pdf-parse, and Mammoth; chunked and embedded using **OpenAI text-embedding-3-small**; stored in **MongoDB Atlas Vector Search** — delivering AI-synthesized, source-cited answers to natural-language queries.
- Designed two **LangGraph agents**: **searchGraph** for library-wide semantic search and **chatGraph** for per-document RAG chat with rolling session memory and summarization — powered by **LangChain.js**, **OpenAI**, and **Anthropic Claude**.
- Implemented **Extractor Factory pattern** for extensible multi-platform ingestion, async background embeddings, Chrome Extension API, Collections, Favourites, **Swagger REST API docs**, and **OAuth 2.0** via NextAuth v5 — deployed on Render with **CI/CD**.

DoodleGlow — Real-time Collaborative Whiteboard

Jul 2025 – Aug 2025

Next.js, TypeScript, Node.js, Express.js, WebSockets, PostgreSQL, Turborepo

- Built a **multi-user collaborative drawing platform** where changes propagate to all connected users in **<100ms** via WebSocket event streams.
- Implemented **infinite scrollable canvas** with optimized dirty-region re-renders — maintaining **60fps** with 10+ concurrent users and tools (freehand, shapes, text, arrows, selection).
- Structured the codebase as a **Turborepo monorepo** with shared TypeScript configs, UI packages, and isolated backend services for scalable development.

Message-Slake — Real-time Chat Application

Jun 2024 – Jul 2024

Node.js, Express.js, React.js, MongoDB, WebSockets, AWS S3, Nodemailer

- Developed a **Slack-inspired messaging app** with real-time communication, threaded replies, and presence indicators using WebSockets.
- Integrated **AWS S3** for secure file/image uploads via pre-signed URLs and **Nodemailer** for asynchronous email notifications.
- Designed **stateless JWT authentication** with room-based WebSocket namespacing for horizontally scalable architecture.

ACHIEVEMENTS

- Solved **500+ DSA problems** on LeetCode — covering arrays, graphs, dynamic programming, and trees.
- Participated in multiple hackathons, building end-to-end web applications and gaining hands-on teamwork experience.

EDUCATION

B.Tech – Computer Science & Engineering | CGPA: 7.2 | **Lovely Professional University, Phagwara, Punjab** Pursuing